

Capstone Challenge – Splunk 101

Splunk SIEM Investigation: Endpoint Compromise & C2 Activity

Scenario: Ryan Adams, a local administrator at a small dental clinic, contacted the MahCyberDefense SOC hotline after suspecting his computer may have been compromised. He explained that around October 15 2025 at 13:00 UTC, his mouse was randomly moving so he became suspicious. As a SOC analyst, your task is to investigate this incident using Splunk. Analyze the available data to determine what occurred, identify any indicators of compromise, and document your findings in a SOC report using the provided format found in the community.

Executive Summary

Following a user report regarding erratic workstation behavior, an investigation was launched into endpoint **FRONTDESK-PC1**. Splunk telemetry analysis (`index="lee-soc"`) confirmed the user interactively executed a suspicious binary (`python.exe`) located in their Music directory.

Upon execution, the malware demonstrated immediate network awareness by establishing outbound communications with an external Command and Control (C2) server, followed by internal RPC connections directly to the organization's Domain Controller (`172.16.0.7`). Furthermore, process telemetry captured PowerShell invoking `schtasks.exe` to establish persistence via a startup scheduled task.

Attack Timeline (15 October 2025)

- **13:00:33 UTC:** User manually executed `python.exe` from the `C:\Users\Ryan.Adams\Music\` directory.
- **13:00:34 UTC:** `python.exe` established an outbound C2 connection to `157.245.46.190:8888`.
- **13:00:34 UTC:** `python.exe` initiated internal RPC connections to the DC (Ports 135 & 49669).
- **13:04:59 UTC:** PowerShell invoked `schtasks.exe` to establish persistence via the `PythonUpdate` task.

Evidence & Technical Findings

1. Process Execution Analysis (T1059)

To gain a high-level overview of malicious commands executed on the host, a summary of all process creation events was generated. This confirmed the execution of `python.exe` and highlighted the exact command line arguments used to establish persistence later in the attack chain.

Splunk Query:

Splunk SPL

1. `index="lee-soc" host="FRONTDESK-PC1" sourcetype="sysmon" EventCode=1 | stats count by _time, Image, CommandLine`

2025-10-15 12:59:29	C:\Users\Ryan.Adams\AppData\Local\Microsoft\OneDrive\25.179.0914.0003\OneDriveLauncher.exe	"C:\Users\Ryan.Adams\AppData\Local\Microsoft\OneDrive\25.179.0914.0003\OneDriveLauncher.exe"
2025-10-15 12:59:30	C:\Windows\System32\wbem\WmiPrvSE.exe	C:\Windows\system32\wbem\wmiprvse.exe -security:NTLM
2025-10-15 13:00:33	C:\Users\Ryan.Adams\Music\python.exe	"C:\Users\Ryan.Adams\Music\python.exe"

2. Malware Execution & Parent-Child Process Trees (T1204 / T1053.005)

By analyzing the parent-child relationships of executed processes, the initial execution vector and persistence mechanisms were identified. The telemetry proves that the user interactively launched the malware (via `explorer.exe`) and that `powershell.exe` was subsequently used to invoke `schtasks.exe` to create a task running as SYSTEM.

Splunk Query:

Splunk SPL

- index="lee-soc" host="FRONTDESK-PC1" sourcetype="*sysmon*" EventCode=1 | sort _time | table _time, Image, ParentImage, CommandLine

2025-10-15 13:02:15	C:\Windows\System32\mmc.exe	C:\Windows\explorer.exe	"C:\Windows\System32\mmc.exe" "C:\Windows\System32\taskchd.msc" /s
2025-10-15 13:03:57	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\explorer.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
2025-10-15 13:04:00	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\explorer.exe	"powershell.exe" -noexit -command Set-Location -literalPath 'C:\Users\Ryan.Adams\Music'
2025-10-15 13:04:53	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	C:\Windows\System32\Task Scheduler\Task Scheduler.exe	"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
2025-10-15 13:04:59	C:\Windows\System32\Task Scheduler\Task Scheduler.exe	C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe	"C:\Windows\System32\Task Scheduler\Task Scheduler.exe" /create /fo PythonUpdate /tr C:\Users\Ryan.Adams\Music\python.exe /sc mustart /ru SYSTEM //

3. Command-and-Control & Internal Network Activity (T1071)

Filtering for Sysmon network events (`EventCode=3`) tied to the malicious executable surfaced two distinct phases of network activity occurring within one second of execution. The malware established an external connection to a suspected C2 IP and immediately initiated internal RPC communications to the Domain Controller, indicating targeted domain reconnaissance.

- External C2:** Outbound connection to `157.245.46.190` over TCP port 8888.
- Internal DC Interaction:** Internal connections to the Domain Controller (`172.16.0.7`) across RPC ports (135 and 49669).

Splunk Query:

Splunk SPL

- index="lee-soc" host="FRONTDESK-PC1" sourcetype="*sysmon*" EventCode=3 Image="*python.exe"

☐ app ▼	C:\Users\Ryan.Adams\Music\python.exe
☐ creation_time ▼	2025-10-15 13:00:33.972
☐ dest ▼	157.245.46.190
☐ dest_ip ▼	157.245.46.190
☐ dest_port ▼	8888
☐ direction ▼	outbound
☐ dvc ▼	FRONTDESK-PC1.KCD.local
☐ dvc_ip ▼	172.16.0.110
☐ dvc_nt_host ▼	FRONTDESK-PC1

Recommendations

1. **Containment:** Immediately isolate **FRONTDESK-PC1** from the network. Remove the **PythonUpdate** scheduled task and securely delete **python.exe**.
2. **Remediation:** Reset all credentials associated with the compromised user account (**Ryan Adams**). Perform a full endpoint health assessment.
3. **Investigation:** Review Domain Controller logs (**172.16.0.7**) to verify if the RPC activity resulted in unauthorized authentication or lateral movement.
4. **Network Defense:** Block the identified malicious IP address (**157.245.46.190**) at the perimeter firewall.
5. **Policy Tuning:** Implement application control rules (e.g., AppLocker) to block the execution of binaries directly out of user profile paths (such as **C:\Users*\Music**).